

Outputs Explanation

The document goes through every output file produced by the programs in the source code archive and explains what is inside it, how to read it, and what it told us. Figures are described but not shown here; the files themselves sit in the numbered output folders.

tert_3month_de_analysis.py

The same differential expression approach applied to the young 3-month fish. The point of the script is to establish the starting position, before the mutation has had time to do anything.

- **tert3m_raw_counts.csv:** the combined count matrix, built by joining the 13 individual featureCounts files into one table with genes as rows and samples as columns. It is worth keeping because the interaction model later needs the 3-month counts in exactly this form.
- **tert3m_pca.png:** PCA of the 3-month samples coloured by genotype. The mutant, heterozygous and wild-type fish do not separate into groups. Normally an unclear PCA is disappointing, but here it is the result itself, because it says the young mutants look like young wild-types.
- **tert3m_MUT_vs_WT_results.csv,** **tert3m_Het_vs_WT_results.csv,** **tert3m_MUT_vs_Het_results.csv:** the three DE tables, same columns as before. The mutant versus wild-type comparison comes back with essentially nothing significant. This near-empty file is one of the most important results in the project, because it establishes that the phenotype is age-dependent rather than present from the start.
- **volcano_MUT_vs_WT.png,** **volcano_Het_vs_WT.png,** **volcano_MUT_vs_Het.png:** volcano plots for the three comparisons. They are noticeably empty compared with the 18-month volcano, with almost nothing pushed into the significant corners, which is the visual version of the same finding.
- **cross_dataset_overlap_3m_vs_18m.csv:** any genes that are significant in both the 3-month and 18-month DEG lists. Written only when the 18-month table is supplied and an overlap exists, which given the 3-month result is a very short list.